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Acoustic building panels

Abstract

Described herein is an acoustic building panel having a first major exposed surface opposite a second major exposed surface and a side exposed surface extending there between, the acoustic ceiling panel comprising: a body comprising a first major surface opposite a second major surface and a side surface extending between the first and second major surfaces, the body being air-permeable; and an attenuation coating applied to the first major surface of the body; wherein a first portion of the second major exposed surface of the acoustic building panel is formed by the first major surface of the body and a second portion of the second major exposed surface of the acoustic building panel is formed by the attenuation coating.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/169,694, filed on Apr. 1, 2021. The disclosure of the above application is incorporated herein by reference.

BACKGROUND

(1) Various types of ceiling systems have been used in commercial and residential building construction to provide the desired acoustical performance. Noise blocking between rooms is required for a variety of purposes, including speech privacy as well as not bothering the occupants of adjacent rooms. Sound dampening within a single room is also required for a variety of purposes, including improving speech comprehension and decreasing volume levels within a single space.

(2) Previous attempts have been made to improve noise blocking between adjacent rooms. However, such previous attempts either lack noise reducing performance or are limited by the maximum sound attenuation that can be achieved. Thus, there is a need for a new acoustic building panel exhibiting the desired enhanced acoustical properties.

BRIEF SUMMARY

(3) In some embodiments, the present invention is directed to an acoustic building panel having a first major exposed surface opposite a second major exposed surface and a side exposed surface extending there between, the acoustic ceiling panel comprising: a body comprising a first major surface opposite a second major surface and a side surface extending between the first and second major surfaces, the body being air-permeable; and an attenuation coating applied to the second major surface of the body; wherein a first portion of the second major exposed surface of the acoustic building panel is formed by the second major surface of the body and a second portion of the second major exposed surface of the acoustic building panel is formed by the attenuation coating.

(4) Other embodiments of the present invention include an acoustic building panel having a first major exposed surface opposite a second major exposed surface and a side exposed surface extending there between, the acoustic ceiling panel comprising: a body that is air-permeable, the body comprising a first major surface opposite a second major surface and a side surface extending there between, the side surface comprising: a lower edge portion adjacent to the first major surface; and an upper edge portion adjacent to the second major surface; an attenuation coating applied to the lower edge portion; wherein a first portion of the side exposed surface of the acoustic building panel is formed by the upper edge portion of the side surface of the body, and a second portion of

the side exposed surface of the acoustic building panel is formed by the attenuation coating.

(5) Other embodiments of the present invention include an acoustic building panel having a first major exposed surface opposite a second major exposed surface and a side exposed surface extending there between, the acoustic ceiling panel comprising: a body comprising a first major surface opposite a second major surface and a side surface extending between the first and second major surfaces, the body being air-permeable; an attenuation coating applied to the first major surface of the body; a plurality of apertures extending through the attenuation coating into the body; and wherein the second major exposed surface of the acoustic building panel comprises the attenuation coating and the plurality of apertures.

(6) Other embodiments of the present invention include a ceiling system comprising: a ceiling grid comprising a plurality of first members and a plurality of second members, the first and second members intersecting one another to define a plurality of grid openings; a plenary space above the ceiling grid; a room environment below the ceiling grid; and at least one of the aforementioned acoustical building panels mounted to the ceiling grid and positioned within the grid opening; and wherein the second major exposed surface of the acoustical building panel faces the plenary space.

(7) Other embodiments of the present invention include a method of forming an acoustic building panel comprising: a) applying an attenuation coating composition to a second major surface of a body in a discontinuous pattern, the body being air-permeable and comprising a first major surface opposite the second major surface and a side surface extending between the first and second major surfaces, b) drying the attenuation coating composition to form the acoustic building panel; and whereby the discontinuous pattern is such that at least a portion of the second major surface of the body is uncoated by the attenuation coating after step b).

(8) Other embodiments of the present invention include a method of forming an acoustic building panel comprising: a) applying a coating composition to a side surface of a body that is air-permeable, the body having a first major surface opposite a second major surface and the side surface extending there-between, the side surface comprising a lower edge portion adjacent to the first major surface and an upper edge portion adjacent to the second major surface; b) drying the coating composition to form an attenuation coating on the acoustic building panel; and wherein the coating composition applied in step a) such that the coating is present on the lower edge portion and wherein after step b) at least a portion of the upper edge portion is free of the attenuation coating.

(9) Other embodiments of the present invention include a method of forming an acoustic building panel comprising: a) applying an attenuation coating composition to a second major surface of a body, the body comprising a first major surface opposite the second major surface and a side surface extending between the first and second major surfaces, b) drying the attenuation coating composition; c) forming a plurality of apertures into the attenuation coating to form the acoustic building panel; and whereby the acoustic building panel comprises a first major exposed surface opposite a second major exposed surface, wherein the plurality of apertures extend from the second major exposed surface to the body.

(10) Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

- (2) FIG. 1 is a perspective view of an acoustic building panel according to the present invention;
- (3) FIG. 2 is a cross-sectional view of the acoustic building panel of FIG. 1 along line II-II;
- (4) FIG. 3 is a top view of the acoustic building panel of FIG. 1;
- (5) FIG. 4 is a perspective view of an acoustic building panel according to another embodiment of the present invention;
- (6) FIG. 5 is a cross-sectional view of the acoustic building panel of FIG. 4 along line V-V;
- (7) FIG. 6 is a building system according to an alternative embodiment of the present invention;
- (8) FIG. 7 is a close-up view of the building system of FIG. 6 in region VII;
- (9) FIG. 8 is a perspective view of an acoustic building panel according to another embodiment of the present invention;
- (10) FIG. 9 is a cross-sectional view of the acoustic building panel of FIG. 8 along line VIII-VIII;
- (11) FIG. 10 is a close up view of region IX of FIG. 9 according to one embodiment; and
- (12) FIG. 11 is a close up view of region IX of FIG. 9 according to another embodiment.

DETAILED DESCRIPTION

(13) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

(14) As used throughout, ranges are used as shorthand for describing each and every value that is within the range. Any value within the range can be selected as the terminus of the range. In addition, all references cited herein are hereby incorporated by referenced in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

(15) The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top,” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such.

(16) Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the exemplified embodiments. Accordingly, the invention expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

(17) Unless otherwise specified, all percentages and amounts expressed herein and elsewhere in the specification should be understood to refer to percentages by weight. The amounts given are based on the active weight of the material. According to the present application, the term “about” means $\pm 5\%$ of the reference value. According to the present application, the term “substantially free” less than about 0.1 wt. % based on the total of the referenced value.

(18) Referring to FIG. 1, the present invention includes a coated building panel **100** (referred to herein as “building panel”) comprising a first major exposed surface **111** opposite a second major exposed surface **112** and a side exposed surface **113** that extends between the first major exposed surface **111** and the second major exposed surface **112**, thereby defining a perimeter **310** of the ceiling panel **100**.

(19) Referring to FIG. 6, the present invention may further include a ceiling system **1** comprising

one or more of the building panels **100** installed in an interior space, whereby the interior space comprises a plenum space **3** and an active room environment **2**. The plenum space **3** provides space for mechanical lines within a building (e.g., HVAC, plumbing, etc.). The active space **2** provides room for the building occupants during normal intended use of the building (e.g., in an office building, the active space would be occupied by offices containing computers, lamps, etc.).

(20) In the installed state, the building panels **100** may be supported in the interior space by one or more parallel support struts **5**. Each of the support struts **5** may comprise an inverted T-bar having a horizontal flange **31** and a vertical web **32**. The ceiling system **1** may further comprise a plurality of first struts that are substantially parallel to each other and a plurality of second struts that are substantially perpendicular to the first struts (not pictured). In some embodiments, the plurality of second struts intersects the plurality of first struts to create an intersecting ceiling support grid **6**. The plenum space **3** exists above the ceiling support grid **6** and the active room environment **2** exists below the ceiling support grid **6**.

(21) In the installed state, the first major exposed surface **111** of the building panel **100** may face the active room environment **2** and the second major exposed surface **112** of the building panel **100** may face the plenum space **3**. The building panel **100** may be installed according to at least two variations. In a first variation, the building panel **100** is positioned entirely above the horizontal flange **31** of the support struts **5**—as shown in FIG. **6**. In the first variation, at least a portion of the first major surface may be concealed from the active room environment **2** by the horizontal flange **31** because the horizontal flange **31** contacts the first major exposed surface **111**, thereby supporting it in the ceiling system **1**. In the first variation, the entire side exposed surface **113** of the building panel **100** may be concealed from the active room environment **2** by the horizontal flange **311**. The second variation will be described further herein.

(22) Referring now to FIGS. **1** and **2**, the building panel **100** of the present invention may have a panel thickness $t_{sub.0}$ as measured from the first major exposed surface **111** to the second major exposed surface **112**. The panel thickness $t_{sub.0}$ may range from about 12 mm to about 40 mm—including all values and sub-ranges there-between. The building panel **100** may have a length $L_{sub.P}$ ranging from about 30 cm to about 310 cm—including all values and sub-ranges there-between. The building panel **100** may have a width $W_{sub.P}$ ranging from about 10 cm to about 125 cm—including all values and sub-ranges there-between.

(23) The building panel **100** may comprise a body **300** and an attenuation coating **400** applied thereto. In some embodiments, the building panel **100** may further comprise a face coating **200** applied to the body **300**—as discussed further herein. The body **300** comprises a first major surface **301** opposite a second major surface **302** and a body side surface **303** that extends between the first major surface **301** and the second major surface **302**, thereby defining a perimeter **310** of the body **300**. The body **300** may be comprised of a binder and fibers. In some embodiments, the body **300** may further comprise a filler and/or additive.

(24) The attenuation coating **400** may be applied to the first major surface **301** of the body **300**. The face coating **200** may be applied to the second major surface **302** of the body **300**. The body **300** may have a body thickness t_i that extends from the first major surface **301** to the second major surface **302**. The body thickness t_i may range from about 12 mm to about 40 mm—including all values and sub-ranges there-between.

(25) The face coating **200** may comprise a binder—such as a polymeric binder—pigments, and processing additives. The face coating **200** may be present in an amount ranging from about 50 g/m^{sup.2} to about 900 g/m^{sup.2}—including all amounts and sub-ranges there-between. The face coating **200** may comprise an upper surface **212** opposite a lower surface **211**. The face coating **200** may be applied such that the lower surface **211** forms the first major exposed surface **111** of the building panel **100**. The face coating **200** may have a solid's content of about 100 wt. %.

(26) The face coating **200** may be applied in a wet-state—i.e., with the addition of a liquid carrier as a face coating composition. The face coating composition may comprise a solid's content of

about 40 wt. % to about 60 wt. % including all sub-ranges and percentages there-between. In some embodiments, the face coating composition may comprise a solid's content of about 50 wt. %.

(27) Although not shown, the building panel **100** of the present invention may further comprise a non-woven scrim. The non-woven scrim may comprise an upper surface opposite a lower surface. The lower surface of the non-woven scrim may be positioned immediately adjacent to and in direct contact with the second major surface **302** of the body **300**. The face coating **200** may be applied to the non-woven scrim such that the upper surface **212** of the face coating **200** is in direct contact with the upper surface of the non-woven scrim.

(28) The body **300** may be porous, thereby allowing airflow through the body **300** between the first major surface **301** and the second major surface **302**—as discussed further herein. According to the present invention, the term porous refers to the body **300** being porous enough to allow for enough airflow through the body **300** (under atmospheric conditions) for the body **300** and the resulting building panel **100** to function as an acoustic building panel **100** and for the corresponding building system **1** to function as an acoustic building system **1**, which requires properties related to noise reduction and sound attenuation properties—as discussed further herein.

(29) Specifically, the body **300** may have a porosity ranging from about 60% to about 98% —including all values and sub-ranges there between. In a preferred embodiment, the body **300** may have a porosity ranging from about 75% to 95%—including all values and sub-ranges there between.

(30) According to the embodiments where the body **300** is formed from binder and fibers, porosity may be calculated by the following:

$$\% \text{ Porosity} = [V.\text{sub.Total} - (V.\text{sub.Binder} + V.\text{sub.F} + V.\text{sub.Filler})] / V.\text{sub.Total}$$

(31) Where V.sub.Total refers to the total volume of the body **300** defined by the first major surface **301**, the second major surface **302**, and the side surfaces **303** of the body **300**. V.sub.Binder refers to the total volume occupied by the binder in the body **300**. V.sub.F refers to the total volume occupied by the fibrous component in the body **300**. V.sub.Filler refers to the total volume occupied by the filler and/or pigment in the body **300**. Thus, the % porosity represents the amount of free volume within the body **300**.

(32) The body **300** of the present invention may exhibit sufficient airflow for the body **300**—and resulting coated building panel **100**—to have the ability to reduce the amount of reflected sound in an active room environment **2**. The reduction in amount of reflected sound in an active room environment **2** is expressed by a Noise Reduction Coefficient (NRC) rating as described in American Society for Testing and Materials (ASTM) test method C423. This rating is the average of sound absorption coefficients at four ½ octave bands (250, 500, 1000, and 2000 Hz), where, for example, a system having an NRC of 0.90 has about 90% of the absorbing ability of an ideal absorber. A higher NRC value indicates that the material provides better sound absorption and reduced sound reflection.

(33) The body **300** of the present invention exhibits an NRC of at least about 0.5. In a preferred embodiment, the body **300** of the present invention may have an NRC ranging from about 0.60 to about 0.99—including all value and sub-ranges there-between.

(34) In addition to reducing the amount of reflected sound in a single active room environment **2**, the body **300** may also be able to exhibit superior sound attenuation—which is a measure of the sound reduction between an active room environment **2** and a plenary space **3**. The ASTM has developed test method E1414 to standardize the measurement of airborne sound attenuation between room environments **2** sharing a common plenary space **3**. The rating derived from this measurement standard is known as the Ceiling Attenuation Class (CAC). Ceiling materials and systems having higher CAC values have a greater ability to reduce sound transmission through the plenary space **3**—i.e. sound attenuation function. The body **300** of the present invention may exhibit a CAC value of 30 or greater.

(35) Non-limiting examples of binder that may form the body **300** may include a starch-based

polymer, polyvinyl alcohol (PVOH), a latex, polysaccharide polymers, cellulosic polymers, protein solution polymers, an acrylic polymer, polymaleic anhydride, epoxy resins, or a combination of two or more thereof. Non-limiting examples of filler may include powders of calcium carbonate, limestone, titanium dioxide, sand, barium sulfate, clay, mica, dolomite, silica, talc, perlite, polymers, gypsum, wollastonite, expanded-perlite, calcite, aluminum trihydrate, pigments, zinc oxide, or zinc sulfate.

(36) Non-limiting examples of fibers that may form the body **300** may include organic fibers, inorganic fibers, or a blend thereof. Non-limiting examples of inorganic fibers mineral wool (also referred to as slag wool), rock wool, stone wool, and glass fibers. Non-limiting examples of organic fiber include fiberglass, cellulosic fibers (e.g. paper fiber—such as newspaper, hemp fiber, jute fiber, flax fiber, wood fiber, or other natural fibers), polymer fibers (including polyester, polyethylene, aramid—i.e., aromatic polyamide, and/or polypropylene), protein fibers (e.g., sheep wool), and combinations thereof.

(37) Referring now to FIGS. **1-3**, the acoustic building panel **100** further comprises an attenuation coating **400** applied to the body **300**. The attenuation coating **400** may be applied to the first major surface **301** of the body **300**—as discussed in greater detail herein.

(38) The attenuation coating **400** may comprise a polymer binder. The polymeric binder may be present in an amount ranging from about 1 wt. % to about 20 wt. % based on the total weight of the dry-state attenuation coating **400**—including all percentages and sub-ranges there-between. Non-limiting examples of binder may include a starch-based polymer, polyvinyl alcohol (PVOH), a latex, polysaccharide polymers, polyvinyl acetate, cellulosic polymers, protein solution polymers, an acrylic polymer, polymaleic anhydride, epoxy resins, or a combination of two or more thereof.

(39) The attenuation coating may comprise a filler. The filler may be present in an amount ranging from about 30 wt. % to about 99 wt. % based on the total weight of the dry-state attenuation coating **400**—including all percentages and sub-ranges there-between. In a preferred embodiment, the filler may be present in an amount ranging from about 50 wt. % to about 99 wt. % based on the total weight of the dry-state attenuation coating **400**—including all percentages and sub-ranges there-between. Non-limiting examples of filler may include pigments, powders of calcium carbonate, including limestone, titanium dioxide, sand, barium sulfate, clay, mica, dolomite, silica, talc, perlite, polymers, gypsum, wollastonite, glass, expanded-perlite, calcite, aluminum trihydrate, pigments, zinc oxide, or zinc sulfate.

(40) In a non-limiting example, the attenuation coating **400** may be applied in the wet-state to the first air-permeable body **300** by spray, roll, curtain coating, screen printing, extrusion coating, or dip application. The attenuation coating **400** may comprise a liquid carrier in the wet-state that is present in an amount ranging from about 20 wt. % to about 60 wt. % based on the total weight of the wet-state attenuation coating—including all percentages and sub-ranges there-between. The attenuation coating **400** may have a solids content in the wet-state that ranges from about 40 wt. % to about 80 wt. % based on the total weight of the wet-state attenuation coating—including all percentages and sub-ranges there-between.

(41) The attenuation coating **400**—in the dry state, i.e., a solids content of about 100 wt. % —may be present atop the first major surface **301** of the body **300** in an amount ranging from about 100 g/m.sup.2 to about 600 g/m.sup.2—including all amounts and sub-ranges there-between. In some embodiments, the attenuation coating **400** may be present atop the first major surface **301** of the body **300** in an amount ranging from about 90 g/m.sup.2 to about 500 g/m.sup.2—including all amounts and sub-ranges there-between.

(42) Once applied, the combination of the attenuation coating **400** and the body **300** form the acoustical panel **100**—whereby the acoustical panel **100** exhibits enhanced attenuation properties due to the presence of the attenuation coating **400**—as discussed further herein.

(43) The second major exposed surface **112** of the building panel **100** may be formed by both of the attenuation coating **400** and the first major surface **301** of the body **300**. Stated otherwise, the

second major exposed surface **112** of the building panel **100** may comprise the attenuation coating **400** and the first major surface **301** of the body **300**.

(44) The attenuation coating **400** may comprise an upper surface **402** opposite a lower surface **401**. The lower surface **401** of the attenuation coating may face the first major surface **301** of the body **300**. The second major exposed surface **112** of the building panel **100** may comprise the upper surface **402** of the attenuation coating **400**.

(45) The attenuation coating **400** may be applied in a plurality of attenuation regions **410** atop the first major surface **301** of the body **300**. Each of the attenuation regions **410** may be a discrete region that is entirely separated by adjacent ones of the attenuation regions **410** by a separation distance $D_{sub.1}$. The separation distance $D_{sub.1}$ may be a non-zero, positive value. The separation distance $D_{sub.1}$ may range from about 2 mm to about 22 mm—including all amounts and sub-ranges there-between.

(46) Each of the plurality of attenuation regions **410** may have a polygonal shape—for example, but not limited to, a rectangle. In other embodiments, the attenuation regions **410** may have a non-polygonal shape—such as a circle, oval, or the like.

(47) In a non-limiting embodiment, each of the attenuation regions **410** may be an elongated polygonal shape that extend substantially parallel to the body **300** along the direction of the width $W_{sub.P}$ of the building panel **100**. In other non-limiting embodiments, each of the discrete regions may be an elongated polygonal shape that extend substantially parallel to the body **300** along the direction of the length $L_{sub.P}$ of the building panel **100**.

(48) The plurality of attenuation regions **410** may include at least two attenuation regions **410** applied atop the first major surface **301** of the body **300**. In some embodiments, the plurality of discrete regions may include at least three attenuation regions **410** applied atop the first major surface **301** of the body **300**. In some embodiments, the plurality of attenuation regions **410** may include at least four attenuation regions **410** applied atop the first major surface **301** of the body **300**.

(49) Each of the attenuation regions **410** may have a length $AR_{sub.L}$ and a width $AR_{sub.W}$. The length $AR_{sub.L}$ of each attenuation region **410** may range from about 250 mm to about 1200 mm—including all distances and sub-ranges there-between. The width $AR_{sub.W}$ of each attenuation region **410** may range from about 28 mm to about 280 mm—including all distances and sub-ranges there-between. A ratio of the length $AR_{sub.L}$ and width $AR_{sub.W}$ of each attenuation region **410** may range from about 40:1 to about 5:1—including all ratios and sub-ranges there-between.

(50) The building panel **100** may further comprise a plurality of offset regions **120** located between adjacent attenuation regions **410**. The offset regions **120** may have a width that is substantially equal to the separation distance $D_{sub.1}$ and a length that is substantially equal to the length $AR_{sub.L}$ of each attenuation region **410**.

(51) A ratio of the width $AR_{sub.W}$ of each attenuation region **410** to the separation distance $D_{sub.1}$ may range from about 15:1 to about 4:1—including all ratios and sub-ranges there-between.

(52) A ratio of the building panel **100** width $W_{sub.P}$ and the separation distance $D_{sub.1}$ may range from about 15:1 to about 3:2—including all ratios and sub-ranges there-between. A ratio of the building panel **100** length $L_{sub.P}$ and the separation distance $D_{sub.1}$ may range from about 600:1 to about 50:1—including all ratios and sub-ranges there-between.

(53) Each of the outermost attenuation regions **410** may be offset and located inwards from the perimeter **310** of the body **300** by a perimeter region **130**. The perimeter region **130** may span a perimeter distance $D_{sub.2}$ that ranges from about 0 mm to about 15 mm—including all amounts and sub-ranges there-between.

(54) A ratio of the width $W_{sub.P}$ of the building panel **100** and the perimeter distance $D_{sub.2}$ of the perimeter region **130** may range from about 1:18 to about 1:4—including all ratios and sub-ranges there-between. A ratio of the length $L_{sub.P}$ of the building panel **100** and the perimeter

distance D.sub.2 of the perimeter region **130** may range from about 1:4 to about 1:2—including all ratios and sub-ranges there-between.

(55) A ratio of the length L.sub.P of the building panel **100** to the length AR.sub.W of each attenuation region **410** may range from about 1:1 to about 3:2—including all ratios and sub-ranges there-between. A ratio of the width W.sub.P of the building panel **100** and the width AR.sub.W of each attenuation region **410** may range from about 500:1 to about 6:1—including all ratios and sub-ranges there-between.

(56) The second major exposed surface **112** of the building panel **100** may have an overall surface area. The first major surface **301** of the body **300** may also have an overall surface area that is substantially equal to the overall surface area of the second major exposed surface **112** of the building panel. The overall surface area of the second major exposed surface **112** of the building panel **100** may be substantially equal to the product of the building panel length L.sub.P and the building panel width W.sub.P (i.e., $L_{sub.P} \times W_{sub.P} = \text{overall surface area}$). The overall surface area of the second major exposed surface **112** may be substantially equal to the surface area located within the perimeter **310** of the body **300** on the first major surface **301**.

(57) Each of the plurality of attenuation regions **410** may occupy a fraction of the overall surface area of the second major exposed surface **112** of the building panel **100**—whereby the summation of the surface areas of the plurality of attenuation regions **410** is a first surface area. Stated otherwise, the attenuation coating **400** occupies a first surface area on the building panel **100**—whereby the first surface area is the summation of each individual surface area of each one of the plurality of attenuation regions **410**.

(58) The first surface area may be less than the overall surface area of the second major exposed surface **112** of the building panel **100**. In some embodiments, the first surface area is equal to about 80% to about 99% of the overall surface area of second major exposed surface **112** of the building panel **100**—including all percentages and sub-ranges there-between. In some embodiments, the first surface area is equal to about 92% to about 98% of the overall surface area of the second major exposed surface **112** of the building panel **100**—including all percentages and sub-ranges there-between.

(59) The plurality of perimeter regions **130** and the plurality of offset regions **120** may further occupy a fraction of the overall surface area of the second major exposed surface **112** of the building panel **100**—whereby the summation of the surface areas of the perimeter regions **130** and the plurality of offset regions **120** is a second surface area.

(60) The second surface area may be calculated as the difference in the overall surface area of the second major exposed surface **112** of the building panel **100** and the first surface area of the plurality of attenuation regions **410**.

(61) The second surface area may be the fraction of the second major exposed surface **112** of the building panel **100** that is formed by the first major surface **301** of the body **300**. Stated otherwise, the second surface area is equal to the portion of the first major surface **301** of the body **300** that is exposed to form a portion of the second major exposed surface **112** of the building panel **100**.

(62) The second surface area may be less than the overall surface area. In some embodiments, the second surface area is equal to about 1% to about 20% of the overall surface area of the second major exposed surface **112** of the building panel **100**—including all percentages and sub-ranges there-between. In some embodiments, the second surface area is equal to about 2% to about 8% of the overall surface area of the second major exposed surface **112** of the building panel **100**—including all percentages and sub-ranges there-between. The first surface area and the second surface area may sum to be substantially equal to the overall surface area.

(63) A ratio of the first surface area to the second surface area may ranging from about 99:1 to about 4:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the first surface area to the second surface area may ranging from about 49:1 to about 11:1—including all ratios and sub-ranges there-between.

(64) As discovered herein, the addition of an attenuation coating **400** in the form of a plurality of discrete attenuation regions **410** allows for a fine-tuning of sound attenuation properties of the building panel **100** and resulting building system **1** without substantial sacrifice to airflow properties necessary for sound reducing characteristics of such panels **100** and related ceiling systems **1**. Specifically, by tailoring the number of individual attenuation regions **410**, the dimensions of such attenuation regions AR.sub.L, AR.sub.W, and the distance from which such attenuation regions **410** may be offset from each other (by the offset distance D.sub.1) and/or offset from the perimeter **310** of the body **300** of the building panel **300** (by the perimeter distance D.sub.2), the attenuation coating **400** of the present invention provides a dynamic approach to better controlling CAC performance of a building panel **100** while also properly counter balancing NRC performance by not completely sacrificing airflow characteristics of the building panel **100** between the first major exposed surface **111** and the second major exposed surface **112**.

(65) Referring now to FIGS. **4** and **5**, a building panel **1100** is illustrated in accordance with another embodiment of the present invention. The building panel **1100** is similar to the building panel **100** except as described herein below. The description of the building panel **100** above generally applies to the building panel **1100** described below except with regard to the differences specifically noted below. A similar numbering scheme will be used for the building panel **1100** as with the building panel **100** except that the 1000-series of numbers will be used.

(66) The building panel **1100** may comprise a first major exposed surface **1111** opposite a second major exposed surface **1112** and a side exposed surface **1113** extending there-between. The side exposed surface **1113** may comprise a first portion **1113a** and a second portion **1113b**. The first portion **1113a** of the side exposed surface **1113** may be located immediately adjacent to the first major exposed surface **1111** of the building panel **1100**, and the second portion **1113b** of the side exposed surface **1113** may be located immediately adjacent to the second major exposed surface **1112** of the building panel **1100**.

(67) The building panel **1000** may comprise a body **1300** and an attenuation coating **1400** applied thereto. In some embodiments, the building panel **1000** may further comprise a face coating **1200** applied to the body **1300**—as discussed further herein.

(68) Although not shown, the building panel **1100** of the present invention may further comprise a non-woven scrim. The non-woven scrim may comprise an upper surface opposite a lower surface. The lower surface of the non-woven scrim may be positioned immediately adjacent to and in direct contact with the second major surface **1302** of the body **1300**. The face coating **1200** may be applied to the non-woven scrim such that the upper surface **1212** of the face coating **1200** is in direct contact with the upper surface of the non-woven scrim.

(69) The body **1300** comprises a first major surface **1301** opposite a second major surface **1302** and a body side surface **1303** that extends between the first major surface **1301** and the second major surface **1302**. The body side surface **1303** may comprise an upper edge portion **1303a** and a lower edge portion **1303b**. The upper edge portion **1303a** may be located immediately adjacent to the first major surface **1301** of the body **1300** and the lower edge portion **1303b** may be located immediately adjacent to the second major surface **1302** of the body **1300**. The lower edge portion **1303b** may extend from the second major surface **1302** of the body **1300** directly to the upper edge portion **1303a**, and the upper edge portion **1303a** may extend directly to the first major surface **1301** of the body **1300**.

(70) The first portion **1113a** of the side exposed surface **1113** of the building panel **1100** may comprise the upper edge portion **1303a** of the body **1303**. Stated otherwise, the first portion **1113a** of the side exposed surface **1113** of the building panel **1100** may be formed by the upper edge portion **1303a** of the body **1303**. A portion of the upper edge portion **1303a** of the body **1303** may form the first portion **1113a** of the side exposed surface **1113** of the building panel **1100**. In some embodiments, the first portion **1113a** of the side exposed surface **1113** of the building panel **1100** may be substantially free of attenuation coating **1400**.

- (71) The attenuation coating **1400** may be applied to the body **1300** such that the second portion **1113b** of the side exposed surface **1113** of the building panel **1100** comprises the attenuation coating **1400**. Stated otherwise, the attenuation coating **1400** may be applied to the body **1300** such that the second portion **1113b** of the side exposed surface **1113** of the building panel **1100** is formed by the attenuation coating **1400**.
- (72) The side exposed surface **1113** may have an overall height that is substantially equal to the panel thickness $t_{sub.0}$. The first portion **1113a** of the side exposed surface **1113** may have a first height $H_{sub.1}$. The first height $H_{sub.1}$ of the first portion **1113a** may range from about 2 mm to about 20 mm—including all heights and subranges therein. The first height $H_{sub.1}$ of the first portion **1113a** may be equal to about 5% of the panel thickness $t_{sub.0}$ to about 50% of the panel thickness $t_{sub.0}$ —including all percentages and sub-ranges therein.
- (73) The second portion **1113b** of the side exposed surface **1113** may have a second height $H_{sub.2}$. The second height $H_{sub.2}$ of the second portion **1113b** may range from about 10 mm to about 45 mm—including all heights and subranges therein. The second height $H_{sub.2}$ of the second portion **1113b** may be equal to about 50% to about 95% of the panel thickness $t_{sub.0}$ —including all percentages and sub-ranges therein.
- (74) The summation of the first height $H_{sub.1}$ of the first portion **1113a** and the second height $H_{sub.2}$ of the second portion **1113b** may be substantially equal to the overall height of the side exposed surface **1113**.
- (75) The upper edge portion **1303a** of the body side surface **1303** may have a third height $H_{sub.3}$. The third height $H_{sub.3}$ of the upper edge portion **1303a** may range from about 4 mm to about 22 mm—including all heights and subranges therein. The third height $H_{sub.3}$ of the upper edge portion **1303a** may be equal to about 12% of the panel thickness $t_{sub.0}$ to about 60% of the panel thickness $t_{sub.0}$ —including all percentages and sub-ranges therein.
- (76) The lower edge portion **1303b** of the body side surface **1303** may have a fourth height $H_{sub.4}$. The fourth height $H_{sub.4}$ of the lower edge portion **1303b** may range from about 8 mm to about 43 mm—including all heights and subranges therein. The fourth height $H_{sub.4}$ of the lower edge portion **1303b** may be equal to about 40% to about 90% of the panel thickness $t_{sub.0}$ —including all percentages and sub-ranges therein.
- (77) The summation of the third height $H_{sub.3}$ of the upper edge portion **1303a** and the fourth height $H_{sub.4}$ of the lower edge portion **1303b** may be substantially equal to the overall height of the side exposed surface **1113**.
- (78) The lower edge portion **1303b** of the body side surface **1303** may comprise a tegular edge profile **1108**. The tegular edge profile **1108** may comprise a vertical wall **1108a** and a horizontal ceiling **1108b**, whereby the vertical wall **1108a** extends from the first exposed major surface **1111** of the building panel **1100** to the horizontal ceiling **1108b** and the horizontal ceiling **1108** extends outward from the vertical wall **1108a**.
- (79) The upper edge portion **1303a** may comprise a bevel edge profile **1109** (also referred to as a “bevel”). The bevel **1109** may extend at an oblique angle relative to both of the second exposed major surface **1112** and the side exposed surface **1113** of the building panel **1100**. In a non-limiting example, the bevel **1109** is oriented at an angle ranging from about 30° to about 60°—including all angles and sub-ranges there-between—relative to one of the second exposed major surface **1112** and the side exposed surface **1113** of the building panel **1100**.
- (80) The tegular edge profile **1108** may overlap with first portion **1113a** of the side exposed surface **1113**. In some embodiments, the tegular edge profile **1108** may fully overlap with first portion **1113a** of the side exposed surface **1113**. The bevel edge profile **1109** may overlap with second portion **1113b** of the side exposed surface **1113**. In some embodiments, the bevel edge profile **1109** may only partially overlap with second portion **1113b** of the side exposed surface **1113**.
- (81) The bevel edge profile **1109** may be formed into the upper edge portion **1303a** of the body side surface **1303**. The bevel edge profile **1109** may overlap with the upper edge profile **1303a** of the

body side surface **1303**. The bevel edge profile **1109** may fully overlap with the upper edge profile **1303a** of the body side surface **1303**. The bevel edge profile **1109** may extend a height that is substantially equal to the third height H.sub.3 of the upper edge profile **1303b**.

(82) The tegular edge profile **1108** may be formed into the lower edge profile **1303b** of the body side surface **1303**. The tegular edge profile **1108** may overlap with the lower edge profile **1303b** of the body side surface **1303**. The tegular edge profile **1108** may fully overlap with the lower edge profile **1303b** of the body side surface **1303**. The tegular edge profile **1108** may extend a height that is less than 100% fraction of the fourth height H.sub.4 of the upper edge profile **1303b**.

(83) The attenuation coating **1400** may be applied to the body side surface **1303** such that the attenuation coating **1400** spans along the body side surface **1303** from the first exposed major surface **1111** to a height that is equal to the second height H.sub.2 of the second portion **1113b** of the side exposed surface **1113**.

(84) A ratio of the second height H.sub.2 of the second portion **1113b** of the side exposed surface **1113** to the first height H.sub.1 of the first portion **1113a** of the side exposed surface **1113** may range from about 1:50 to about 1:1—including all ratios and sub-ranges therein.

(85) A ratio of the fourth height H.sub.4 of the lower edge portion **1303b** of body side surface **1303** to the third height H.sub.3 of the upper edge portion **1303a** of the body side surface **1303** may range from about 25:1 to about 1:1—including all ratios and sub-ranges therein.

(86) The attenuation coating **1400** may be applied to the lower edge portion **1303b** of the body **1300**. The attenuation coating **1400** may be applied such that the attenuation coating **1400** coats substantially the entirety of the lower edge portion **1303b** of the body **1300**. The attenuation coating **1400** may be applied such that the attenuation coating **1400** continuously coats the entirety of the lower edge portion **1303b** of the body **1300**.

(87) The attenuation coating **1400** may be applied to the upper edge portion **1303a** of the body **1300**. The attenuation coating **1400** may be applied partially to the upper edge portion **1303a** of the body **1300**. The attenuation coating **1400** may be applied partially to the upper edge portion **1303a** of the body **1300** such that the first portion **1113a** of the side exposed surface **1113** remains substantially free of attenuation coating **1400**. The attenuation coating **1400** may be applied partially to the upper edge portion **1303a** of the body **1300** such that the first portion **1113a** of the side exposed surface **1113** is formed by the upper edge portion **1303a** of the body **1300**.

(88) The second height H.sub.2 of the second portion **1113b** of the side exposed surface **1113** may be greater than the fourth height H.sub.4 of the lower edge portion **1303b** of the body side surface **1303** of the body **130**. A ratio of the second height H.sub.2 of the second portion **1113b** of the side exposed surface **1113** to the fourth height H.sub.4 of the lower edge portion **1303b** of the body side surface **1303** of the body **130** may range from about 1:1 to about 4:1—including all ratios and sub-ranges therein.

(89) The third height H.sub.3 of the upper edge portion **1303a** of the body side surface **1303** of the body **130** may be greater than the first height H.sub.1 of the first portion **1113a** of the side exposed surface **1113**. A ratio of the third height H.sub.3 of the upper edge portion **1303a** of the body side surface **1303** to the first height H.sub.1 of the first portion **1113a** of the side exposed surface **1113** may range from about 1:1 to about 10:1—including all ratios and sub-ranges therein.

(90) The attenuation coating **1400**—in the dry state—i.e., a solid's content of about 100 wt. %—may be present on the body side surface **1303** in an amount ranging from about 30 g/m.sup.2 to about 170 g/m.sup.2—including all amounts and sub-ranges there-between.

(91) The attenuation coating **1400** may be applied as an attenuation coating composition in a wet-state—i.e., with the addition of a liquid carrier as a face coating composition. The attenuation coating composition may comprise a solid's content of about 60 wt. % to about 85 wt. %—including all sub-ranges and percentages there-between. In some embodiments, the attenuation coating composition may comprise a solid's content of about 75 wt. %.

(92) As discovered herein, the attenuation coating **1400** applied such that the second portion **1113b** of

the side exposed surface **1113** of the building panel **1110** comprises the attenuation coating **1400** while the first portion **1113a** of the side exposed surface **1113** of the building panel **1100** is substantially free of the attenuation coating **1400** allows for a fine-tuning of sound attenuation properties of the building panel **1100** and resulting building system **1** without substantial sacrifice to airflow properties necessary for sound reducing characteristics of such panels **1100** and related ceiling systems **1**. Specifically, by controlling the attenuation coating **1400** present on the side exposed surface **1113**, whereby the side exposed surface comprises a bevel **1109**, the attenuation coating **1400** of the present invention provides a dynamic approach to better controlling CAC performance of a building panel **1100** while also properly counter balancing NRC performance by not completely sacrificing airflow characteristics of the building panel **1100** between the active room environment **2** and the plenary space **3** for installed building systems **1**.

(93) Referring now to FIGS. **8-10**, a building panel **2100** is illustrated in accordance with another embodiment of the present invention. The building panel **2100** is similar to the building panel **100** and the building panel **1100** except as described herein below. The description of the building panel **100** and the building panel **1100** above generally applies to the building panel **2100** described below except with regard to the differences specifically noted below. A similar numbering scheme will be used for the building panel **2100** as with the building panel **100** and the building panel **1100** except that the 2000-series of numbers will be used.

(94) The acoustic building panel **2100** comprises an attenuation coating **2400** applied to the body **2300**. The attenuation coating **2400** may be applied to the first major surface **2301** of the body **2300**—as discussed in greater detail herein. In some embodiments, the acoustic building panel **2100** may further comprise a face coating **2200** applied to the second major surface **2302** of the body **2300**.

(95) Although not shown, the building panel **2100** of the present invention may further comprise a non-woven scrim. The non-woven scrim may comprise an upper surface opposite a lower surface. The lower surface of the non-woven scrim may be positioned immediately adjacent to and in direct contact with the second major surface **2302** of the body **2300**. The face coating **2200** may be applied to the non-woven scrim such that the upper surface **2212** of the face coating **2200** is in direct contact with the upper surface of the non-woven scrim.

(96) The attenuation coating **2400**—in the dry-state—may be atop the first major surface **2301** of the body **2300** in an amount ranging from about 20 g/m.² to about 36 g/m.²—including all amounts and sub-ranges there-between. In some embodiments, the attenuation coating **2400** may be atop the first major surface **2301** of the body **2300** in an amount ranging from about 25 g/m.² to about 30 g/m.²—including all amounts and sub-ranges there-between.

(97) The attenuation coating **2400** may comprise an upper surface **2402** opposite a lower surface **2401**. The lower surface **2401** of the attenuation coating **2400** may face the first major surface **2301** of the body **2300**. The second major exposed surface **2112** of the building panel **2100** may comprise the upper surface **2402** of the attenuation coating **2400**.

(98) The second major exposed surface **2112** of the building panel **2100** may comprise the attenuation coating **2400** and a plurality of apertures **2500**. Each of the apertures **2500** may be an open pathway extending from the second major exposed surface **2112** of the building panel **2100** through the attenuation coating **2400** to the body **2300**. In some embodiments, each of the apertures **2500** may be an open pathway extending continuously from the second major exposed surface **2112** of the building panel **2100** through the attenuation coating **2400** to the body **2300**.

(99) In some embodiments, each of the apertures **2500** may be an open pathway extending from the second major exposed surface **2112** of the building panel **2100** from the upper surface **2402** of the attenuation coating **2400** through the lower surface **2401** of the attenuation coating **2400** to the body **2300**. In some embodiments, each of the apertures **2500** may be an open pathway extending continuously from the second major exposed surface **2112** of the building panel **2100** from the upper surface **2402** of the attenuation coating **2400** through the lower surface **2401** of the

attenuation coating **2400** to the body **2300**.

(100) In some embodiments, each of the apertures **2500** may be an open pathway extending from the second major exposed surface **2112** of the building panel **2100** from the upper surface **2402** of the attenuation coating **2400** through the lower surface **2401** of the attenuation coating **2400** and through the first major surface **2301** of the body **2300**. In some embodiments, each of the apertures **2500** may be an open pathway extending continuously from the second major exposed surface **2112** of the building panel **2100** from the upper surface **2402** of the attenuation coating **2400** through the lower surface **2401** of the attenuation coating **2400** and through the first major surface **2301** of the body **2300**.

(101) Each of the plurality of apertures **2500** may comprise a first end **2501** opposite a second end **2502** and an aperture wall **2503** extending between the first end **2501** and the second end **2502**. The aperture wall **2503** may extend continuously between the first end **2501** and the second end **2502** of each aperture **2500**, thereby defining a hollow channel that allows for the open pathway extending between the first end **2501** and the second end **2502** of each of the plurality of apertures **2500**.

(102) Each of the plurality of apertures **2500** may comprise a central axis A-A that extends between the first end **2501** and the second end **2502**. According to some embodiments, the central axis A-A may be substantially parallel to the aperture **2503**.

(103) The aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the second major exposed surface **2112** of the building panel **2100**. The aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the first major surface **2301** of the body **2300**. The second end **2502** of each of the plurality of apertures **2500** may form a floor, whereby the floor is substantially parallel to the second major exposed surface **2112** of the building panel **2100**. The second end **2502** of each of the plurality of apertures **2500** may form a floor, whereby the floor is substantially parallel to the first major surface **2301** of the body **2300**.

(104) Each of the plurality of apertures **2500** may have an aperture depth $D_{\text{sub.3}}$ as measured by the distance spanning between the first end **2501** and the second end **2502**. The aperture depth $D_{\text{sub.3}}$ may range from about 0.0625 inches to about 0.25 inches—including all depths and sub-ranges there-between. In some embodiments, the aperture depth $D_{\text{sub.3}}$ may range from about 0.0625 inches to about 0.188 inches—including all depths and sub-ranges there-between. In some embodiments, the aperture depth $D_{\text{sub.3}}$ may be about 0.125 inches.

(105) Each of the plurality of apertures **2500** may have an aperture diameter $D_{\text{sub.4}}$ as measured by the distance between opposite sides of the aperture wall **2503**. The aperture diameter $D_{\text{sub.4}}$ may range from about 0.02 inches to about 0.375 inches—including all depths and sub-ranges there-between. In some embodiments, the aperture diameter $D_{\text{sub.4}}$ may range from about 0.03 inches to about 0.03 inches—including all diameters and sub-ranges there-between. In some embodiments, the aperture diameter $D_{\text{sub.4}}$ may be about 0.375 inches.

(106) A ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may range from about 1:6 to about 8:1—including all ratios and sub-ranges there-between.

(107) In some embodiments, the aperture depth $D_{\text{sub.3}}$ may be larger than the aperture diameter $D_{\text{sub.4}}$. A ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may range from about 1.5:1 to about 8:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may range from about 2:1 to about 6:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may range from about 3:1 to about 5:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may be about 4:1.

(108) In some embodiments, the aperture depth $D_{\text{sub.3}}$ may be smaller than the aperture diameter $D_{\text{sub.4}}$. A ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may range from about 1:6 to about 1:1.1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the aperture depth $D_{\text{sub.3}}$ to the aperture diameter $D_{\text{sub.4}}$ may range from about 1:4

to about 1:1.1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the aperture depth D.sub.3 to the aperture diameter D.sub.4 may range from about 1:4 to about 1:2—including all ratios and sub-ranges there-between.

(109) The aperture wall **2503** may comprise a first portion **2503a** and a second portion **2503b**. The first portion **2503a** may extend from the first end **2501** of each of the plurality of apertures **2500** to the second portion **2503b**. The second portion **2503b** may extend from the first portion **2503a** to the second end **2502** of each of the plurality of apertures **2500**.

(110) The first portion **2503a** of the aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the second major exposed surface **2112** of the building panel **2100**. The second portion **2503b** of the aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the second major exposed surface **2112** of the building panel **2100**. The first portion **2503a** of the aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the first major surface **2301** of the body **2300**. The second portion **2503b** of the aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the first major surface **2301** of the body **2300**. The first portion **2503a** of the aperture wall **2503** may extend vertically in a direction that is substantially parallel to the central axis A-A, and the second portion **2503b** of the aperture wall **2503** may extend vertically in a direction that is substantially parallel to the central axis A-A.

(111) The first portion **2503a** of the aperture wall **2503** may extend a first aperture depth D.sub.3A. The second portion **2503b** of the aperture wall **2503** may extend a second aperture depth D.sub.3B. The summation of the first aperture depth D.sub.3A and the second aperture depth D.sub.3B may be substantially equal to the aperture depth D.sub.3. The first aperture depth D.sub.3A may be equal to about 20% to about 80% of the aperture depth D.sub.3—including all percentages and sub-ranges there-between. The second aperture depth D.sub.3B may be equal to about 30% to about 70% of the aperture depth D.sub.3—including all percentages and sub-ranges there-between.

(112) The first portion **2503a** of the aperture wall **2503** may be formed by the attenuation coating **2400**. The second portion **2503b** of the aperture wall **2503** may be formed by the body **2300**.

(113) Each of the plurality of apertures **2500** may be an open pathway extending from the first end **2501** located at the second major exposed surface **2112** of the building panel **2100** to the second **2502** located inside of the body **2300**. In such embodiments, the open pathway may be defined by an aperture wall **2503** that includes a first portion **2503a** formed by the attenuation coating **2400** and extending the first aperture depth D.sub.3A that is substantially equal to the thickness of the attenuation coating **2400** as measured between the upper surface **2402** and the lower surface **2401** of the attenuation coating **2400**. In such embodiments, the open pathway may continue from the first portion **2503b** of the aperture wall **2503** along the second portion **2503b** of the aperture wall **2503** that extends to the second aperture depth D.sub.3B, which may be the total depth the aperture **2500** extends from the first major surface **2301** of the body **2300** into the core **2308** of the body **2300**.

(114) The second major exposed surface **2112** of the building may comprise the upper surface **2402** of the attenuation coating **2400** and the first end **2501** of each of the plurality of apertures **2500**.

(115) When viewing second major exposed surface **2112** of the building panel **2000** in a direction extending from the second major exposed surface **2112** to the first major exposed surface **2111**, the upper surface **2402** of the attenuation coating **2400** may be visible as well as a portion of the core **2308** of the body **2300** that is located at least one of the second portion **2503b** of the aperture wall **2503** and/or the second end **2502** of each of the plurality of apertures **2500** via the open pathway of each aperture **2500**.

(116) The second major exposed surface **2112** of the building panel **2100** may have an overall surface area as defined by the perimeter **2310** of the body **2300**, the attenuation coating **2400** may occupy a first surface area, and the first end **2501** of each of the plurality of apertures **2400** may occupy a second surface area, wherein the summation of the first surface area and the second

surface area is substantially equal to the overall surface area of the second major exposed surface **2112** of the building panel **2100**.

(117) Each of the plurality of apertures **2500** may be offset by an aperture separation distance $D_{sub.5}$ —as measured by the distance spanning between adjacent most apertures **2500** without an intervening aperture **2500**. The aperture separation distance $D_{sub.5}$ may range from about 0.125 inches to about 2.0 inches—including all distances and sub-ranges there-between. In some embodiments, the aperture separation distance $D_{sub.5}$ may range from about 0.25 inches to about 0.75 inches—including all distances and sub-ranges there-between. In some embodiments, the aperture separation distance $D_{sub.5}$ may be about 0.25 inches. In some embodiments, the aperture separation distance $D_{sub.5}$ may be about 0.375 inches. In some embodiments, the aperture separation distance $D_{sub.5}$ may be about 0.5 inches. In some embodiments, the aperture separation distance $D_{sub.5}$ may be about 0.625 inches. In some embodiments, the aperture separation distance $D_{sub.5}$ may be about 0.75 inches.

(118) The plurality of apertures **2500** may be present on the building panel **2100** in an aperture density ranging from about 20 aperture/ft.^{sup.2} to about 9100 aperture/ft.^{sup.2}—including all densities and sub-ranges there-between. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density ranging from about 20 aperture/ft.^{sup.2} to about 600 aperture/ft.^{sup.2}—including all densities and sub-ranges there-between. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density ranging from about 30 aperture/ft.^{sup.2} to about 350 aperture/ft.^{sup.2}—including all densities and sub-ranges there-between. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density ranging from about 200 aperture/ft.^{sup.2} to about 600 aperture/ft.^{sup.2}—including all densities and sub-ranges there-between.

(119) In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density of about 37 aperture/ft.^{sup.2}. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density of about 57 aperture/ft.^{sup.2}. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density of about 48 aperture/ft.^{sup.2}. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density of about 89 aperture/ft.^{sup.2}. In some embodiments, the plurality of apertures **2500** may be present on the building panel **2100** in an aperture density of about 350 aperture/ft.^{sup.2}.

(120) In some embodiments, the aperture diameter $D_{sub.4}$ may range from about 0.03 inches to about 0.063 inches, the aperture separation distance $D_{sub.5}$ may range from about 0.1 inches to about 0.15 inches, and having an aperture density ranging from about 8,900 aperture/ft.^{sup.2} to about 9,100 aperture/ft.^{sup.2}.

(121) In some embodiments, the aperture diameter $D_{sub.4}$ may range from about 0.03 inches to about 0.063 inches, the aperture separation distance $D_{sub.5}$ may range from about 0.4 inches to about 0.6 inches, and having an aperture density ranging from about 400 aperture/ft.^{sup.2} to about 600 aperture/ft.^{sup.2}.

(122) In some embodiments, the aperture diameter $D_{sub.4}$ may range from about 0.03 inches to about 0.063 inches, the aperture separation distance $D_{sub.5}$ may range from about 0.9 inches to about 1.1 inches, and having an aperture density ranging from about 100 aperture/ft.^{sup.2} to about 140 aperture/ft.^{sup.2}.

(123) In some embodiments, the aperture diameter $D_{sub.4}$ may range from about 0.03 inches to about 0.063 inches, the aperture separation distance $D_{sub.5}$ may range from about 1.9 inches to about 2.1 inches, and having an aperture density ranging from about 20 aperture/ft.^{sup.2} to about 30 aperture/ft.^{sup.2}.

(124) In some embodiments, the aperture diameter $D_{sub.4}$ may range from about 0.05 inches to about 0.08 inches, the aperture separation distance $D_{sub.5}$ may range from about 0.1 inches to

about 0.15 inches, and having an aperture density ranging from about 8,900 aperture/ft.² to about 9,100 aperture/ft.².

(125) In some embodiments, the aperture diameter D.sub.4 may range from about 0.05 inches to about 0.08 inches, the aperture separation distance D.sub.5 may range from about 0.4 inches to about 0.6 inches, and having an aperture density ranging from about 400 aperture/ft.² to about 600 aperture/ft.².

(126) In some embodiments, the aperture diameter D.sub.4 may range from about 0.05 inches to about 0.08 inches, the aperture separation distance D.sub.5 may range from about 0.9 inches to about 1.1 inches, and having an aperture density ranging from about 100 aperture/ft.² to about 140 aperture/ft.².

(127) In some embodiments, the aperture diameter D.sub.4 may range from about 0.05 inches to about 0.08 inches, the aperture separation distance D.sub.5 may range from about 1.9 inches to about 2.1 inches, and having an aperture density ranging from about 20 aperture/ft.² to about 30 aperture/ft.².

(128) In some embodiments, the aperture diameter D.sub.4 may range from about 0.1 inches to about 0.15 inches, the aperture separation distance D.sub.5 may range from about 0.4 inches to about 0.6 inches, and having an aperture density ranging from about 400 aperture/ft.² to about 600 aperture/ft.².

(129) In some embodiments, the aperture diameter D.sub.4 may range from about 0.1 inches to about 0.15 inches, the aperture separation distance D.sub.5 may range from about 0.9 inches to about 1.1 inches, and having an aperture density ranging from about 100 aperture/ft.² to about 140 aperture/ft.².

(130) In some embodiments, the aperture diameter D.sub.4 may range from about 0.1 inches to about 0.15 inches, the aperture separation distance D.sub.5 may range from about 1.9 inches to about 2.1 inches, and having an aperture density ranging from about 20 aperture/ft.² to about 30 aperture/ft.².

(131) In some embodiments, the aperture diameter D.sub.4 may range from about 0.3 inches to about 0.4 inches, the aperture separation distance D.sub.5 may range from about 0.4 inches to about 0.6 inches, and having an aperture density ranging from about 400 aperture/ft.² to about 600 aperture/ft.².

(132) In some embodiments, the aperture diameter D.sub.4 may range from about 0.3 inches to about 0.4 inches, the aperture separation distance D.sub.5 may range from about 0.9 inches to about 1.1 inches, and having an aperture density ranging from about 100 aperture/ft.² to about 140 aperture/ft.².

(133) In some embodiments, the aperture diameter D.sub.4 may range from about 0.3 inches to about 0.4 inches, the aperture separation distance D.sub.5 may range from about 1.9 inches to about 2.1 inches, and having an aperture density ranging from about 20 aperture/ft.² to about 30 aperture/ft.².

(134) Each of the plurality of apertures **2500** may have an open pathway with a cross-sectional shape that is circular, ovular, or polygonal. The plurality of apertures **2500** may have a substantially uniform distribution across the second major exposed surface **2112** of the building panel **2100**. In other embodiments, the plurality of apertures **2500** may have a non-uniform distribution across the second major exposed surface **2112** of the building panel **2100**.

(135) Referring now to FIG. **11**, a building panel **2100i** is illustrated in accordance with another embodiment of the present invention. The building panel **2100i** is similar to the building panel **2100** except as described herein below. The description of the building panel **2100** generally applies to the building panel **2100i** described below except with regard to the differences specifically noted below. A similar numbering scheme will be used for the building panel **2100i** as with the building panel **2100** except that the 2000i-series of numbers will be used.

(136) The second major exposed surface **2112i** of the building panel **2100i** may comprise the

attenuation coating **2400i** and a plurality of apertures **2500i**. Each of the plurality of apertures **2500i** may comprise a first end **2501i** opposite a second end **2502i** and an aperture wall **2503i** extending between the first end **2501i** and the second end **2502i**.

(137) The aperture wall **2503i** may extend continuously between the first end **2501i** and the second end **2502** of each aperture **2500i**, thereby defining a hollow channel that allows for the open pathway extending between the first end **2501i** and the second end **2502i** of each of the plurality of apertures **2500i**.

(138) Each of the plurality of apertures **2500** may comprise a central axis Ai-Ai that extends between the first end **2501i** and the second end **2502i**. The aperture wall **2503i** may comprise a first portion **2503ai** and a second portion **2503bi**. The first portion **2503ai** may extend from the first end **2501i** of each of the plurality of apertures **2500i** to the second portion **2503bi**. The second portion **2503bi** may extend from the first portion **2503ai** to the second end **2502i** of each of the plurality of apertures **2500i**.

(139) The aperture wall **2503** may extend vertically in a multi-direction. The first portion **2503ai** of the aperture wall **2503** may extend vertically in a direction that is substantially orthogonal to the first major surface **2301** of the body **2300** and the second portion **2503bi** of the aperture wall **2503** may extend vertically in a direction that is oblique to the first major surface **2301i** of the body **2300i**.

(140) The first portion **2503ai** of the aperture wall **2503** may extend in a direction that is substantially parallel to the central axis Ai-Ai and the second portion **2503bi** of the aperture wall **2503** may extend vertically in a direction that is oblique to the central axis Ai-Ai.

(141) The second portion **2503bi** may extend inward toward the central axis Ai-Ai and terminate at the second end **2502i**. The second portion **2503bi** of the aperture wall **2503i** may form an inverted conical shape. The second end **2502i** may be an apex of the conical shape or may form a floor of an inverted truncated conical shape.

(142) The building panel **2100** according to this embodiment may be formed by applying an attenuation coating composition—in the wet-state—to the first major surface **2301** of the body **230** and drying the attenuation coating composition to form an un-perforated attenuation coating.

(143) The building panel **2100** according to this embodiment may be formed by applying the attenuation coating composition—in the wet-state—to the first major surface **2301** of the body **2300** in a continuous manner such that substantially the entirety of the first major surface **2301** of the body **2300** is coated with the attenuation coating composition. Stated otherwise, the attenuation coating composition—in the wet-state—may be applied in a continuous manner to the entirety of the first major surface **2301** of the body **2300** such that substantially the entirety of the first major surface **2301** of the body **2300** is coated with the attenuation coating composition.

(144) Subsequently, the unperforated attenuation coating may be machined to form the plurality of apertures **2500** that extend from the upper surface **2402** of the aperture coating **2400** into the core **2308** of the body **2300**. In a non-limiting example, the plurality of apertures **2500** may be formed by a needle-punching process—whereby the needle-punching may be operated to achieve the final aperture depth D.sub.3, aperture diameter D.sub.4, aperture separation distance D.sub.5, and aperture density. The attenuation coating **2400** of this embodiment may be referred to as a “perforated attenuation coating” **2400** to refer to the fact that the plurality of apertures extend through the attenuation coating **2400** between the lower surface **2401** and the upper surface **2402**.

(145) As discovered herein, the addition of the plurality of apertures **2500** to the building panel **2100** allows for a fine-tuning of sound attenuation properties of the building panel **2100** and resulting building system **1** without substantial sacrifice to airflow properties necessary for sound reducing characteristics of such panels **2100** and related ceiling systems **1**. Specifically, by tailoring the final aperture depth D.sub.3, aperture diameter D.sub.4, aperture separation distance D.sub.5, and aperture density, the perforated attenuation coating **2400** of the present invention provides a dynamic approach to better controlling CAC performance of a building panel **2100** while also

properly counter balancing NRC performance by not completely sacrificing airflow characteristics of the building panel **100** between the first major exposed surface **2111** and the second major exposed surface **2112**.

(146) The building panel **2100** may exhibit an airflow resistance as measured between the first major exposed surface **2111** and the second major exposed surface **2112**—the airflow resistance measured by the following formula:

$$R=(P.\text{sub}.A-P.\text{sub}.ATM)/V$$

(147) Where R is air flow resistance (measured in ohms); P.sub.A is the applied air pressure; P.sub.ATM is atmospheric air pressure; and V is volumetric airflow. The airflow resistance of the building panel **2100** may range from about 5 ohms to about 25 ohms—including all airflow resistances and sub-ranges there-between. In some embodiments, the building panel **2000** may exhibit an airflow resistance as measured between the first major exposed surface **2111** and the second major exposed surface **2112** that ranges from about 8 ohms to about 22 ohms—including all airflow resistances and sub-ranges there-between. In some embodiments, the building panel **2000** may exhibit an airflow resistance as measured between the first major exposed surface **2111** and the second major exposed surface **2112** that ranges from about 10 ohms to about 20 ohms—including all airflow resistance and sub-ranges there-between.

Claims

1. An acoustic building panel having a first major exposed surface opposite a second major exposed surface and a side exposed surface extending there between, the acoustic ceiling panel comprising: a body that is air-permeable, the body comprising a first major surface opposite a second major surface and a side surface extending there between, the side surface comprising: a lower edge portion adjacent to the first major surface; and an upper edge portion adjacent to the second major surface; an attenuation coating applied to the first major surface, the attenuation coating applied in a plurality of discrete attenuation regions, each discrete attenuation region separated from adjacent attenuation regions by at least a portion of the body; wherein each discrete attenuation region is separated from the side surface such that a perimeter on the first major surface is free from the attenuation coating and is defined around the plurality of discrete attenuation regions; and wherein a first portion of the side exposed surface of the acoustic building panel is formed by the upper edge portion of the side surface of the body, and a second portion of the side exposed surface of the acoustic building panel is formed by the lower edge portion.

2. The acoustic building panel according to claim 1, wherein the side exposed surface has an overall height, the first portion has a first height, and the second portion has a second height, wherein the summation of the first height and the second height is substantially equal to the overall height of the side exposed surface.

3. The acoustic building panel according to claim 2, wherein a ratio of the second height to the first height ranges from about 1:1 to about 10:1.

4. The acoustic building panel according to claim 1, wherein the side surface comprises a bevel that is adjacent to the second major surface of the body.

5. The acoustic building panel according to claim 4, wherein the bevel overlaps with the first portion of the side exposed surface of the acoustic building panel and the second portion of the side exposed surface of the acoustic building panel.

6. The acoustic building panel according to claim 5, wherein the bevel fully overlaps with the first portion of the side exposed surface of the acoustic building panel and partially overlaps with the second portion of the side exposed surface of the acoustic building panel.

7. The acoustic building panel according to claim 1, wherein the body is formed of a fibrous material.

8. The acoustic building panel according to claim 7, wherein the fibrous material comprises an

- inorganic fiber selected from one or more of fiberglass, mineral wool, rock wool, and stone wool.
9. The acoustic building panel according to claim 1, wherein the body exhibits an NRC value of at least 0.7.
10. The acoustic building panel according to claim 1, wherein the attenuation coating comprises a polymer binder and a filler.
11. The acoustic building panel according to claim 2, wherein the upper edge portion of the body has a third height and the lower edge portion has a fourth height, and wherein the first height and third height are not equal and wherein the second height and the fourth height are not equal.
12. A ceiling system comprising: a ceiling grid comprising a plurality of first members and a plurality of second members, the first and second members intersecting one another to define a plurality of grid openings; a plenary space above the ceiling grid; a room environment below the ceiling grid; and the acoustical building panel according to claim 1 mounted to the ceiling grid and positioned within the grid opening; wherein the second major exposed surface of the acoustical building panel faces the plenary space.
13. The ceiling system according to claim 12, wherein the side exposed surface of the acoustic building panel at least partially overlaps with the ceiling grid in a horizontal direction.
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